AI-Assisted Application Evaluation

Challenge 2: Manual testing compared with AI-assisted testing

Application under test: ElectraStore (validate.womentechquest.com)

Women Tech Quest 2026, QA Challenge

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1. Scope and approach

Both rounds of testing were done on the same application, ElectraStore, a fictional electrical-appliance retailer. Round 1 was done by hand in the browser. Round 2 used an AI assistant (Claude) that read the live site pages, calculated values from the data on them and compared the results with the rules the site publishes.

What the AI could and could not do. The AI read the rendered text of the search results page and a product page. It could not click, add items to a cart, sign in, or watch how the interface behaves. So it is strong at checking data, calculations and content, and it could not repeat the interactive bugs from Round 1. This limit is stated again in the comparison.

2. Manual findings (Challenge 1)

Four bugs were submitted manually. All four were marked **Counted** by the platform, and the challenge was locked on 10 Oct 2026, 07:01.

#	Bug title	Area	Status
M1	Hyperlink issue: email hyperlink not working	Links / contact	Counted
M2	Appliance image icons on widgets missing	UI / images	Counted
M3	Favourite product count is incorrect after removing a product from favourites	Favourites	Counted
M4	Brand category: on all tabs the Clear Filters button is not working	Filters / category	Counted

The full steps, expected result and actual result for each bug are in the Challenge 1 submission and are not repeated here.

3. AI findings (Challenge 2)

A1. Discount badge is calculated on the sale price, not the list price

Where: search results, product page and related-product cards. **Severity:** Medium, because customers see an inflated discount.

Steps: open the search page for “IcePoint 54L” and read the badge on the product card. Compare it with the two prices shown.

Expected: discount = (list price - sale price) / list price. **Actual:** the badge matches (list price - sale price) / sale price.

Product	List price	Sale price	Badge shown	Correct discount
IcePoint 54L Refrigerator	PKR 104,900	PKR 87,500	-20%	16.6%
CoolLife 240L Double Door	PKR 59,000	PKR 50,000	-18%	15.3%
ChillMate 56L Refrigerator	PKR 213,500	PKR 166,499	-28%	22.0%
CoolLife 16L Refrigerator	PKR 113,499	PKR 98,990	-15%	12.8%
CoolLife Inverter 400L	PKR 116,500	PKR 100,000	-17%	14.2%

The “You save PKR 17,400” line on the product page is correct. Only the percentage is wrong, and the same wrong formula appears on all five products checked, so it is systematic.

A2. Implausible product data

Severity: Low to Medium (data quality; it may mislead customers).

• The IcePoint 54L Refrigerator lists a power consumption of 1200 W. That is very high for a 54-litre unit, and the page also describes it as being among “single and double-door refrigerators with inverter compressors”, which does not fit a 54L model.
• The ChillMate 56L Refrigerator is priced at PKR 166,499 and the CoolLife 16L Refrigerator at PKR 98,990. These prices are far above the PKR 87,500 of the 54L unit and the PKR 50,000 of a 240L double-door, so price does not follow capacity.

A3. Product copy is generic

Severity: Low. The description on the product page repeats the category text (“single and double-door refrigerators…”) instead of describing the specific model. The title says 54L but the copy says it supports double-door refrigerators.

A4. Rules available to test in the app (not yet executed)

The Help Centre states rules that can be turned into test cases. They were **not** run by the AI because the AI could not interact with the app: free delivery at PKR 25,000 or more; maximum 10 units per product (the product page shows “Maximum 10 per order”); delivery date window; password rules; cancellation only before Packed. These are listed as AI-generated test ideas, not as defects.

4. Comparison: manual vs AI

Category	Findings
Found by both	None. The manual bugs (M1 to M4) were interactive or visual, and the AI could only read page text, so it did not repeat them.
Found only manually	M1 email hyperlink, M2 missing appliance icons, M3 favourites count after removal, M4 Clear Filters button on the brand category.
Found only by AI	A1 wrong discount percentage (5 of 5 products checked), A2 implausible specs and prices, A3 generic product copy.
What manual testing missed	The discount calculation. It is easy to miss by eye because 20% looks believable, and it only shows up when the numbers are worked out. The data-quality problems were also not noticed.
What AI missed	State-based and UI bugs: the broken link, missing icons, the favourites counter and the filter button. The AI cannot click or see the rendered screen.

5. Strengths and limitations

	Manual testing	AI-assisted testing
Strengths	Finds UI, interaction and state bugs. Sees what the user sees. Good at judging whether the app feels right.	Fast. Checks arithmetic and data across many items at once. Turns written rules into test cases. Notices patterns such as a wrong formula repeated everywhere.
Limitations	Slow. Easy to miss calculation errors. Coverage depends on how much time the tester has.	Cannot click, sign in or see the screen when it only reads page text. Findings need to be verified by a person. May treat unusual data as wrong when it is a business choice.

6. Conclusion

The two methods found different kinds of defects, and nothing overlapped. Manual testing found four interface and behaviour bugs. The AI found a systematic calculation bug and data-quality issues that a quick manual pass would likely miss. Neither method was enough alone. The best approach is to let a person test the interface and let AI check data, calculations and rule coverage, then have a person confirm every AI finding before it is reported.

Limitation of this report: the AI did not execute the interactive test cases in A4, so those remain suggestions. Running them in the app would be the next step.